Creating A VPC

When creating a VPC you need to have your network mapped out so that everything can go smoothly. Im going to use the same network i used for the homework section:

I statically assigned the IP but when building the actual vpc i will be using DHCP to assign the IP addresses.
AZ1 us-east-1a
Public:
Network: 10.32.1.0/24
IP: 10.32.1.6
Mask: 255.255.255.0

AZ2 us-eaast-1b
Public:
Network: 10.32.2.0/24
IP: 10.32.2.7
Mask: 255.255.255.0

AZ3 us-east-1c
Public:
Network: 10.32.3.0/24
IP: 10.32.3.8
Mask: 255.255.255.0

AZ1 us-east-1a
Private:
Network: 10.32.11.0/24
IP: 10.32.11.11
Mask: 255.255.255.0

AZ2 us-eaast-1b
Private:
Network: 10.32.12.0.0/24
IP: 10.32.12.12
Mask: 255.255.255.0

AZ3 us-east-1c
Private
Network: 10.32.13.0.0/24
IP: 10.32.13.31
Mask: 255.255.255.0

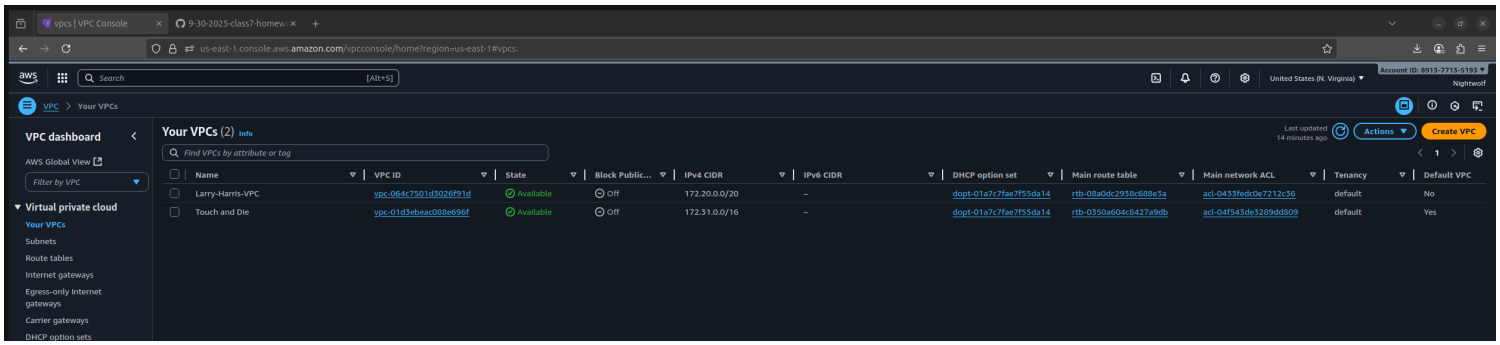
AZ1 us-east-1a
Private
Network: 10.32.111.0.0/24
IP: 10.32.111.41
Mask: 255.255.255.0

AZ2 us-eaast-1b

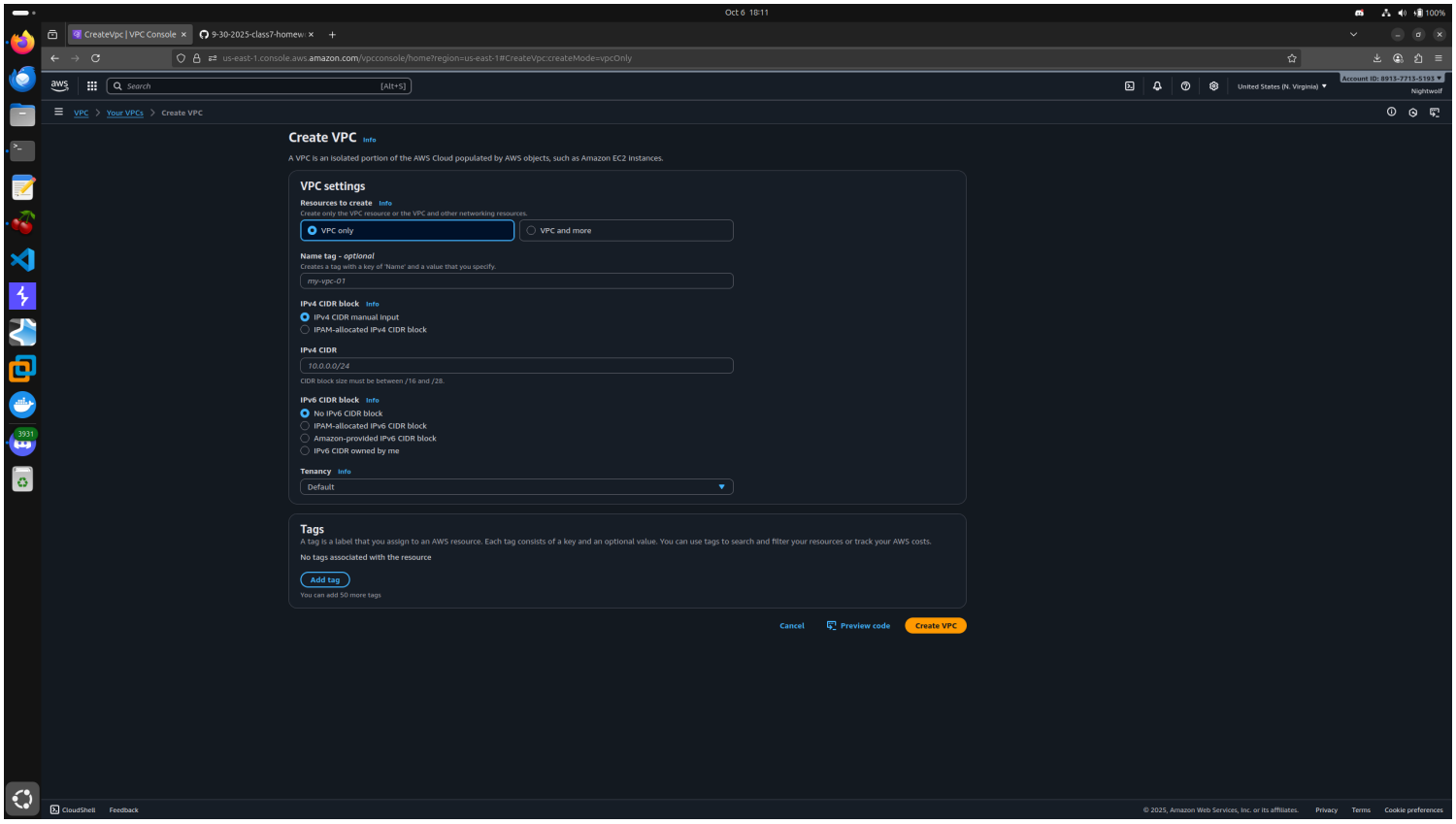
Private
Network: 10.32.112.0.0/24
IP: 10.32.112.51
Mask: 255.255.255.0

AZ3 us-east-1c
Private
Network: 10.32.113.0.0/24
IP: 10.32.113.61
Mask: 255.255.255.0

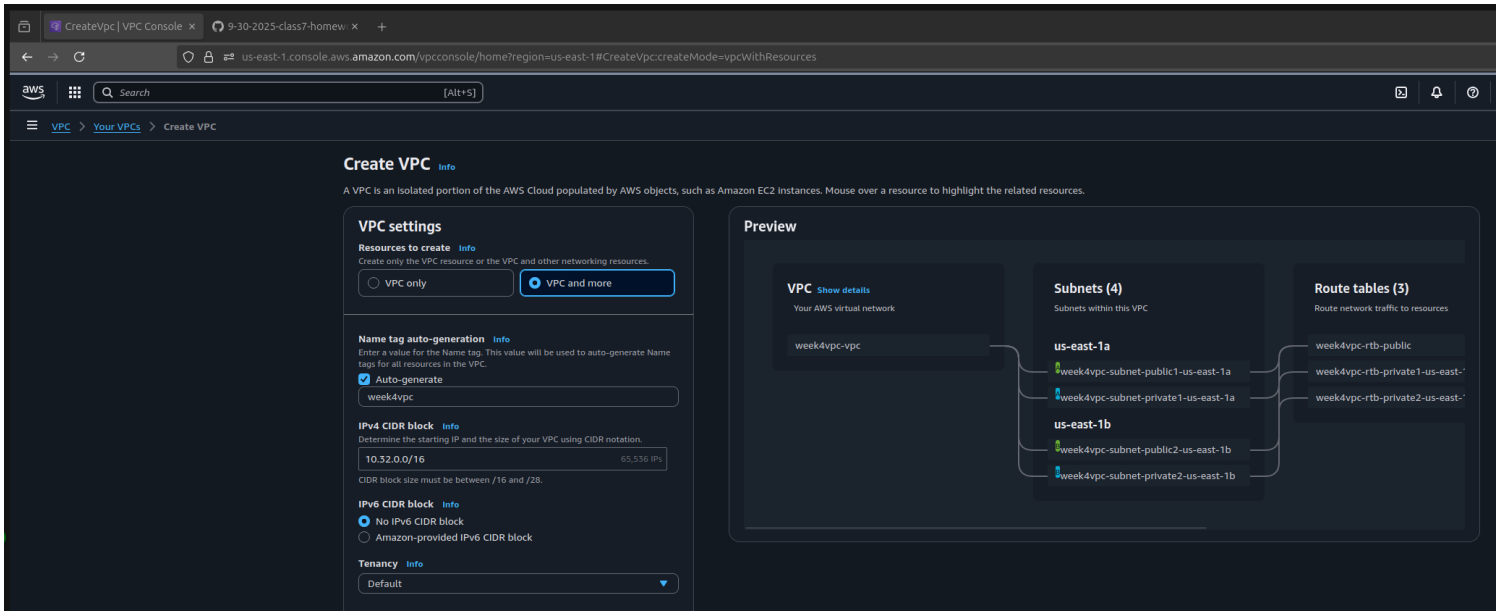
Ok I start of by searcing for VPC and the create VPC



As you can see i currently have 2 VPC's. The default one and one that i created about 5 years ago when i first started learning AWS. I then press the Create VPC button:



Here I choose VPC and more, name it week4vpc, IPV4 CIDER i use 10.32.0.0 as my network states to use.



Number of availabilty zones i am using are 3 with 3 public subnets and 6 private subnets: The subnets are derived from the network:

I then verify before moving forward.
Nat Gateways I am choosing In 1 AZ and VPC endpoints i set to none before creating the vpc:

NAT gateways ($) Info

Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway

None | In 1 AZ | 1 per AZ

VPC endpoints Info

Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None | S3 Gateway

DNS options Info

- [x] Enable DNS hostnames
- [x] Enable DNS resolution

▸ Additional tags

Cancel Preview code Create VPC

I then create the VPC

Create VPC workflow

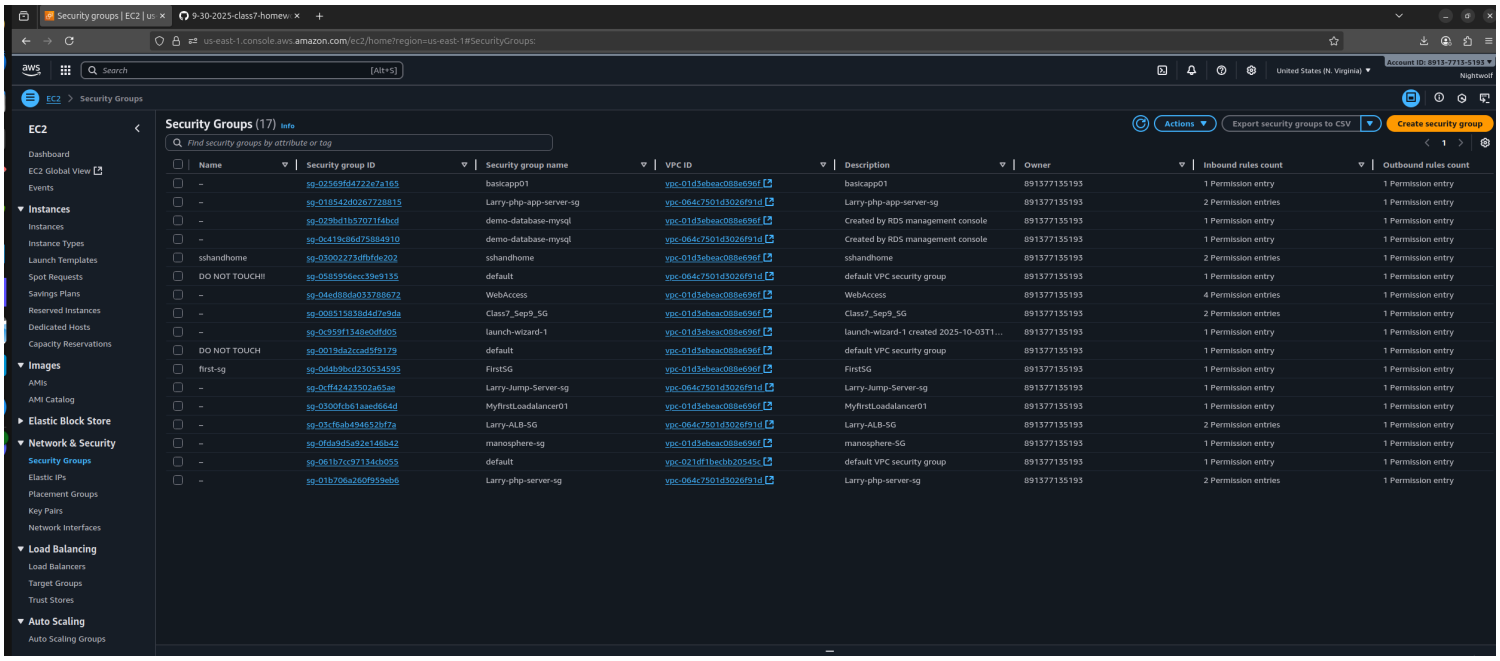
Success

Details

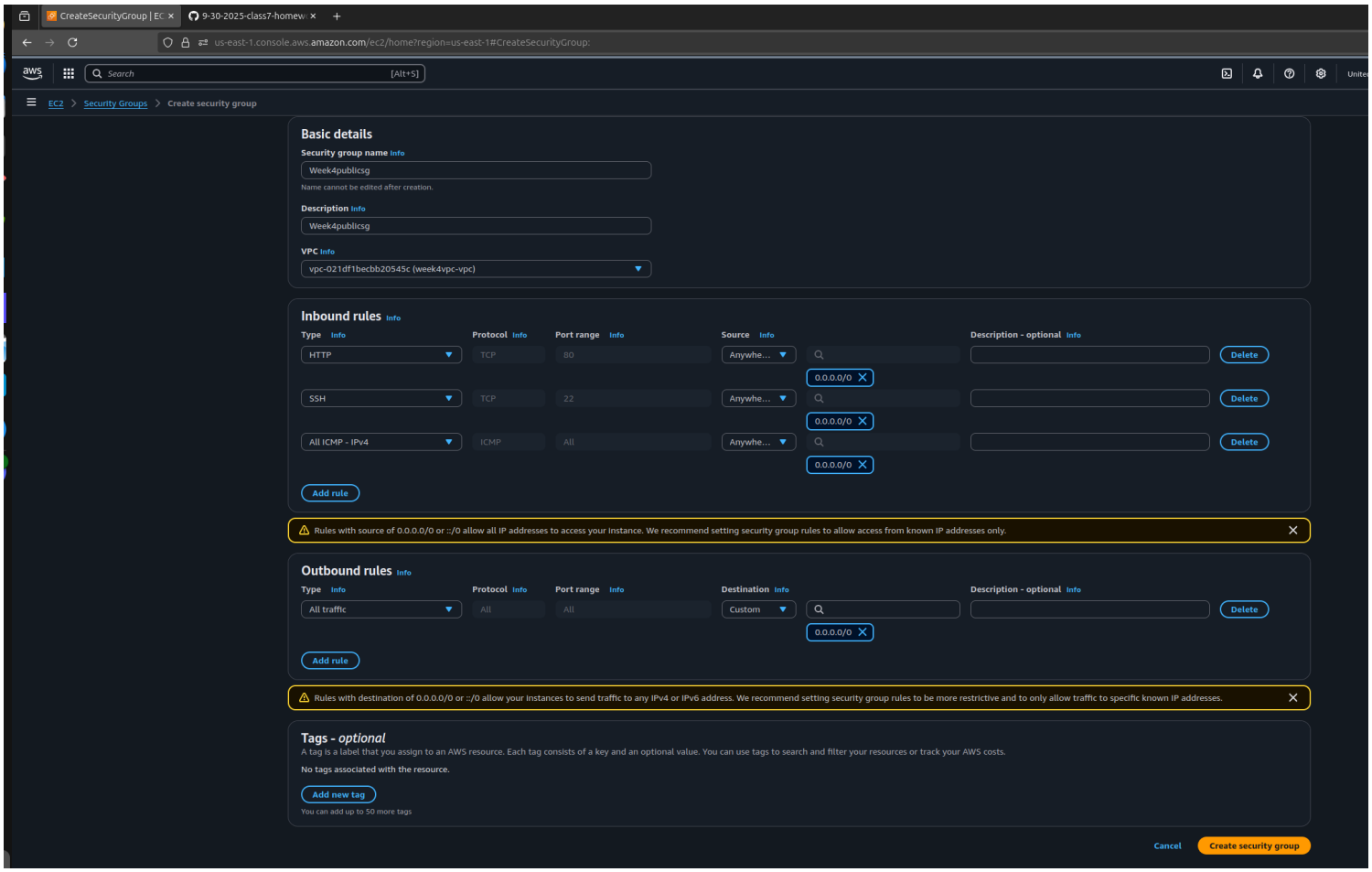
- Create VPC: vpc-021df1becbb20545c
- Enable DNS hostnames
- Enable DNS resolution
- Verifying VPC creation: vpc-021df1becbb20545c
- Create subnet: subnet-0c6342afd6ba9f065
- Create subnet: subnet-076ab797e675350b4
- Create subnet: subnet-0b47296db3730d135
- Create subnet: subnet-064400bf08b4afafc
- Create subnet: subnet-01ea42a7b86c12343
- Create subnet: subnet-07150ba4dff0fdd7b
- Create subnet: subnet-0e87607f819d5aa72
- Create subnet: subnet-0b782c4d08b857395
- Create subnet: subnet-09d8225c5fc6b82f6
- Create internet gateway: igw-015b3e1dcf0e60ee4
- Attach internet gateway to the VPC
- Create route table: rtb-0d082d7ea48754e7a
- Create route
- Associate route table
- Associate route table
- Associate route table
- Create route table: rtb-04a410742c51ad156
- Associate route table
- Create route table: rtb-0fa2314e5f67f3f21
- Associate route table
- Create route table: rtb-090faff71f7f8a74e
- Associate route table
- Create route table: rtb-0f54b3950d56ad192
- Associate route table
- Create route table: rtb-041002261e1ea6774
- Associate route table
- Create route table: rtb-0265b0f506c484298
- Associate route table
- Verifying route table creation

Everything seems to be good!!!
Now I am going to create my security groups for the instances that i am going to build. I am building 1 public and 1 private for testing and documentation purposes.

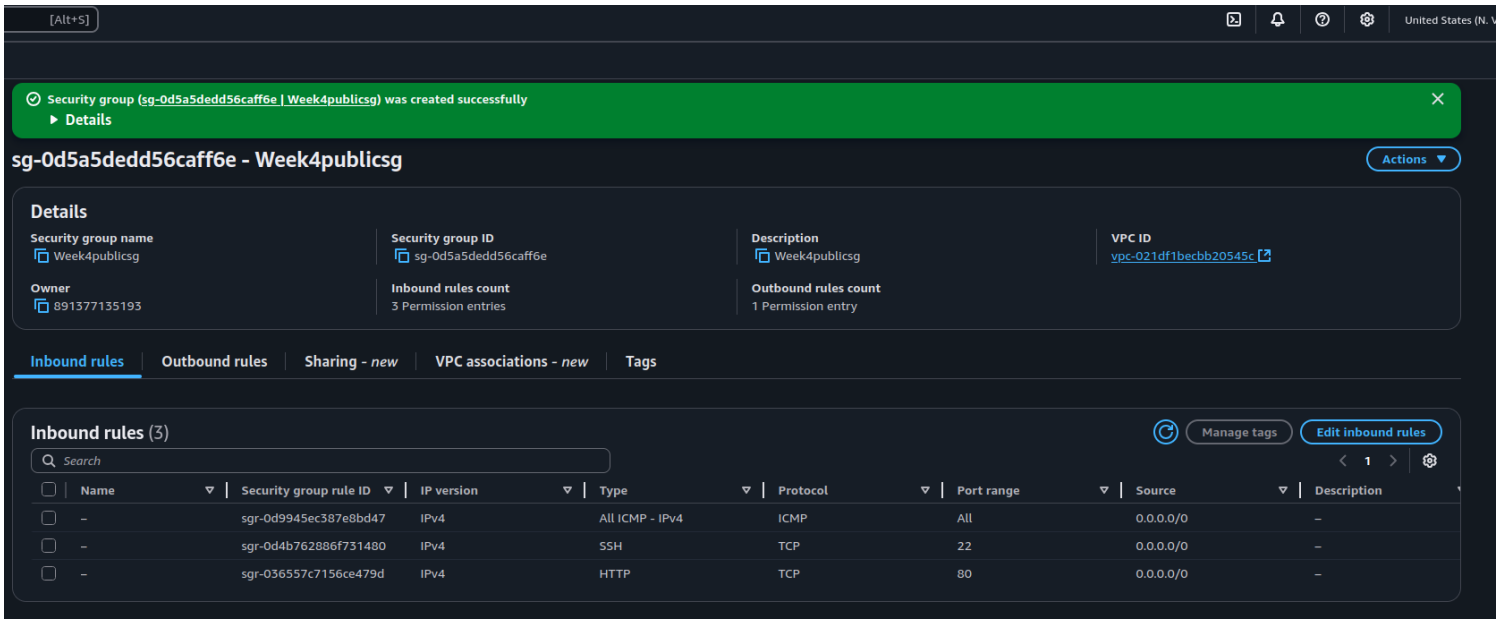
To build my security groups i go to EC2 and under Network & security to security groups



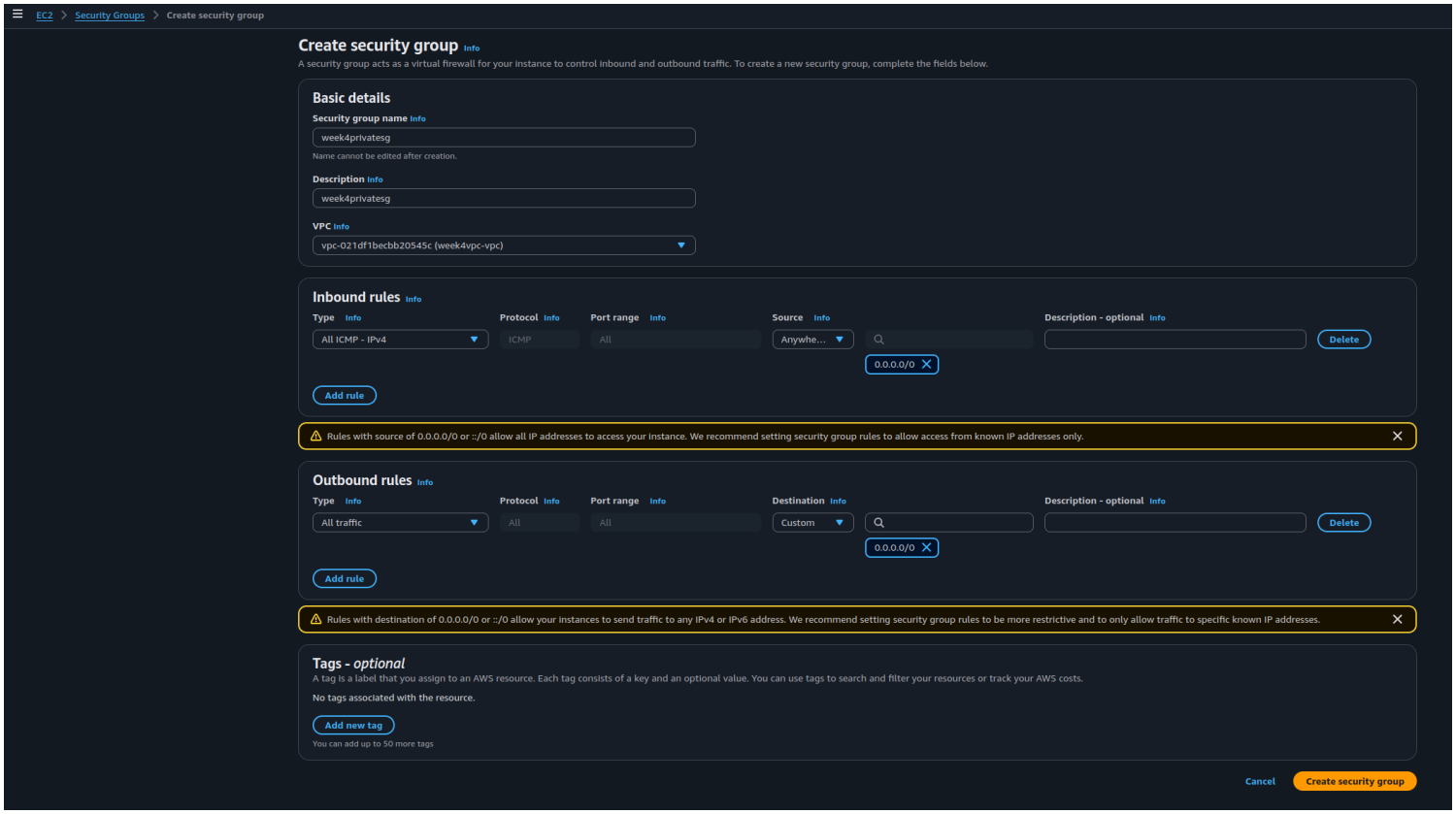
Ok I am creating 2 security groups. 1 for public and 1 for private.
Click the create scurity group orange tab:
I give it a appropriate name and description: week4publicsg,
I change the vpc from default to the one i created whichis week4vpc,
I add inbound rules: HTTP, SSH, ICMP:



I then create the security group:



Now i create the private security group:
For testing purposes i am only enabling ICMP:



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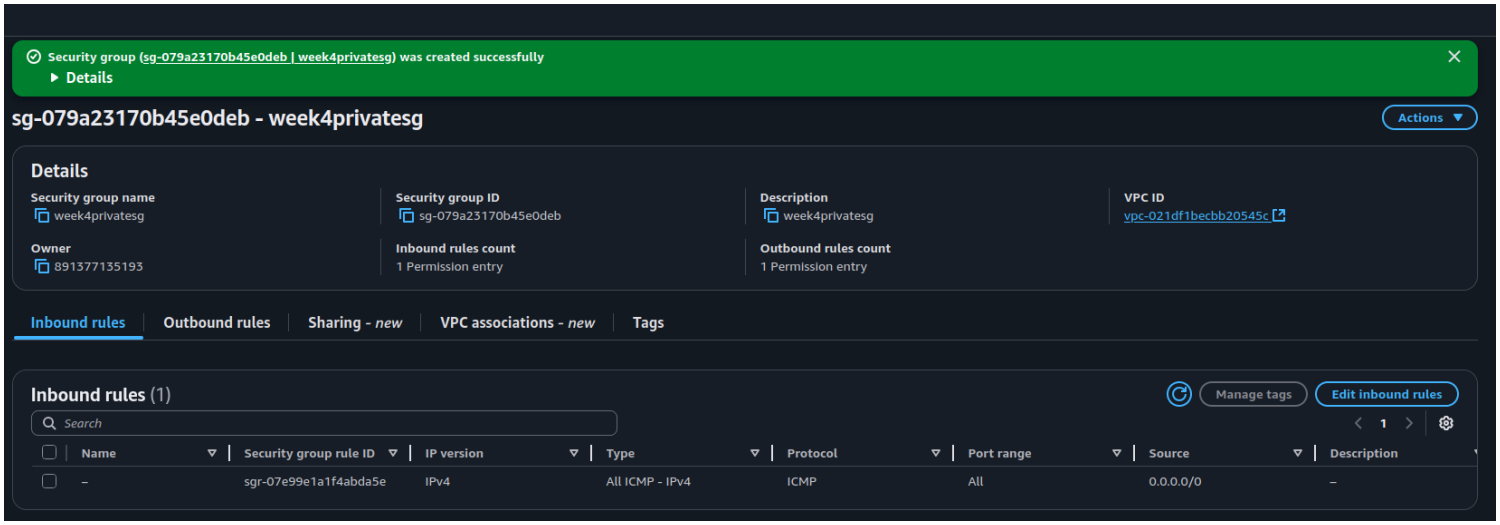
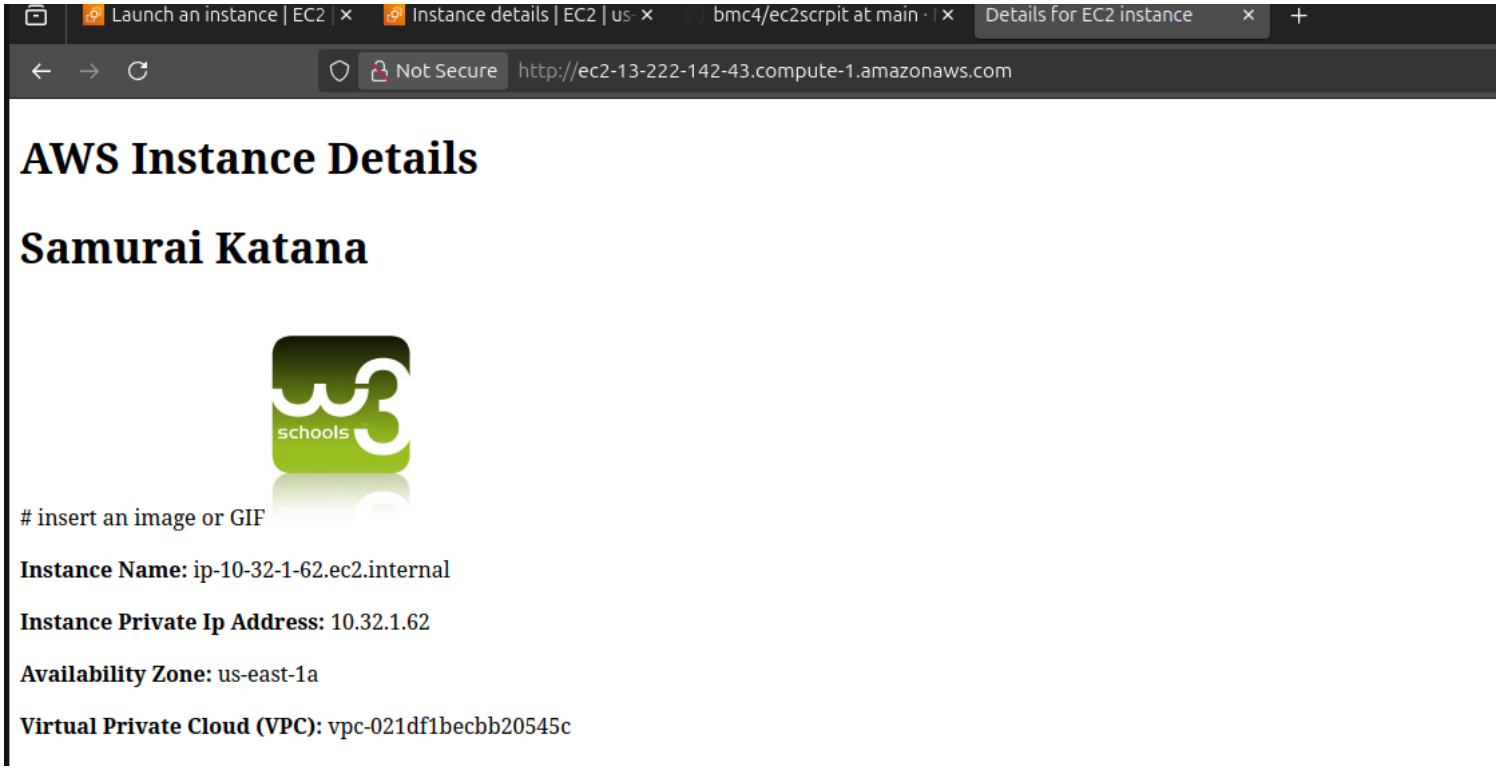


Now i build the 2 instances for testing:

Launch instance,
I name it public test,
Proceed without a key pair,
Under network settings: Change the vpc to the one created and under subnet choose the correct subnet for public:
Enable the auto-assign public ip
select existing security group and choose the security group for the public as this is the public instance,
Go to Advanced Details and then paste the startup script into the user data section:
Launch the instance:

Create the second instance with similar steps:
Launch instance,
name: week4private,

Proceed without keypair,
Edit network settings and change vpc,
choose subnet
enable public assign ip
select the private security group
I pasted the script in the user data in case we wanted to enable http later.

Results from the public server: